RAJALAKSHMI ENGINEERING COLLEGE

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S2

Started on	Monday, 31 August 2026, 10:03 PM
State	Finished
Completed on	Monday, 31 August 2026, 10:04 PM
Time taken	41 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include <stdio.h>

void quickSort(int a[],int low,int high){
    if(low<high){
        int i=low,j=high,pivot=a[low],t;
        while(i<j){
            while(a[i]<=pivot&&i<high)i++;
            while(a[j]>pivot)j--;
            if(i<j){
                t=a[i];a[i]=a[j];a[j]=t;
            }
        }
        a[low]=a[j];a[j]=pivot;
        quickSort(a,low,j-1);
        quickSort(a,j+1,high);
    }
}

int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++)scanf("%d",&a[i]);
    quickSort(a,0,n-1);
    for(int i=0;i<n;i++){
        if(i)printf(" ");
        printf("%d",a[i]);
    }
    return 0;
}

```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔

	Input	Expected	Got	
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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